

Problem-Solution Fit:

Date	12 July 2026
Team Id	XXXXX
Project Name	Opti Crop : Smart Agricultural Product Optimization Engine
Maximum marks	5 marks

Problem-Solution Fit:

1. CUSTOMER SEGMENTS (CS) • Farmers (Small & Large Scale) • Agricultural Researchers • Agribusiness Companies • Government Agricultural Officers	6. CUSTOMER LIMITATIONS (CL) • Limited internet access • Lack of technical knowledge • Budget constraints • Limited access to agricultural experts	5. AVAILABLE SOLUTIONS (AS) Solutions: Traditional farming, agricultural experts, farming websites. Pros: • Easy to access • Expert guidance • Time-consuming • Costly • Less accurate than AI predictions Cons:
2. PROBLEMS / PAINS (PR) • Low crops yield due to improper crop selection. • Lack of real-time soil and weather analysis. • High fertilizer and irrigation costs. • Unpredictable weather affecting crops.	9. PROBLEM ROOT / CAUSE (RC) • Incorrect crop selection. • Climate change. • Poor soil nutrient management. • Lack of scientific farming practices.	7. BEHAVIOR (BH) • Consults local farmers or experts. • Searches online for crop information. • Uses previous farming experience. • Performs soil testing before sowing.

3. TRIGGERS TO ACT (TA) . New farming season begins. • Soil test results are available. • Weather conditions change. • Need to improve productivity	10. YOUR SOLUTION (SL) Smart Agricultural Production Optimization Engine is an AI-based crop recommendation system that analyses soil nutrients, temperature, humidity, rainfall, and pH to recommend the most suitable crop, helping farmers improve	8. CHANNELS OF BEHAVIOR (CH) ONLINE • Agricultural websites • Government portals • YouTube • Mobile apps
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	productivity and make data-driven decisions.	OFFLINE • Agricultural officers • Local markets • Farmer meetings • Training programs
4. EMOTIONS Before: Confused, worried, uncertain about crop selection. After: Confident, satisfied, optimistic about higher yields.		